StormCrews: An Open Simulator of Distribution Restoration After Storms

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***Abstract*—After a storm, how long customers stay without power depends on decisions a utility controls: how many crews it has, where it sends them, which repairs go first and when it asks neighboring utilities for help. StormCrews is an open-source Python simulator of those decisions. It replays 85 real storms of 2018 to 2025 in 189 counties of New England, Pennsylvania and West Virginia, built only from public data, and reports each run in reliability terms: the storm's contribution to SAIDI, customers out beyond 24, 72 and 96 hours, crew-hours and crews on duty. Damage arrives hour by hour as in the outage records, crews travel and repair jobs, and mutual aid joins after a lag. A restoration rule takes a few lines of code and runs on every storm with common random numbers, and a fitted damage model changes the weather of a real storm. Calibrated per storm, with the fleet set from crew counts that utilities reported in public filings, the simulator reproduces observed customer-hours. With its repair parameters predicted from the other storms instead, a single storm's customer-hours miss by 16 % at the median in New England. The inputs, the decisions and the outputs are described, the valid uses are stated, and a weather what-if, a sweep of crews and mutual-aid lag and a comparison of restoration rules illustrate the tool.**

Keywords*—*distribution system restoration, reliability indices, resilience, repair crews, mutual assistance, discrete-event simulation, open-source software

I. Introduction

Severe weather causes the largest outages on distribution grids, and restoring service afterwards is an operations problem with planning consequences [1], [2]. Damage keeps appearing for hours after the first outages. The control room sees customers out by area, not the faults behind them. Repairs take uncertain times, crews lose hours driving, and crews borrowed from other utilities arrive hours or days after the request. A planner who wants to know how much a larger local fleet, a faster mutual-aid agreement or a different dispatch rule would change the reliability indices of a bad season has few tools to ask with.

Fig. 1 shows the problem on a real storm. When Tropical Storm Isaias crossed New England in August 2020, customers out peaked at 1,220 thousand, yet 24 % of the customers who lost power did so after that hour, and nine tenths of the damage had arrived only 17 hours after the peak. A day after the peak 787 thousand customers were still out, and four days after it 201 thousand. The event added 552 minutes to the region's SAIDI, counting major event days.

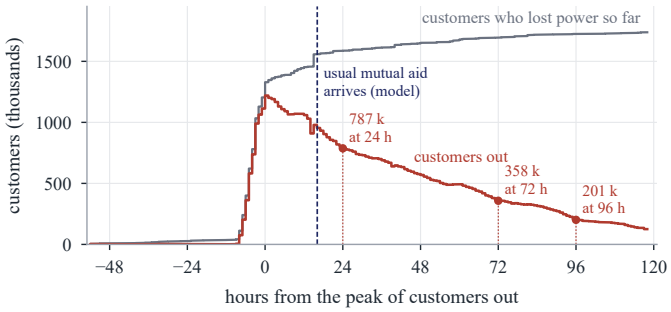


Fig. 1. Damage keeps arriving after the peak and restoration takes days. Tropical Storm Isaias in New England, August 2020: customers out (red) and customers who had lost power so far (gray), EAGLE-I hourly records, with customers still out 24, 72 and 96 hours after the peak. The dashed line marks when the usual mutual aid, requested once 1 % of customers are out, joins in the simulator.

The usual tools answer other questions. Network optimization schedules repairs and switching on a known feeder model [3]. Agent-based and statistical models of restoration time are fitted on one utility's private records [4], [5], so others cannot rerun them. StormCrews sits between the two. It needs no network model, only the customers out in each area hour by hour, and it is calibrated and checked against real storms from public records, so anyone can rerun every number.

The contribution is the tool itself, an open and tested simulator of crews, mutual aid and restoration rules on 85 real storms whose outputs are the reliability quantities planners and regulators already track.

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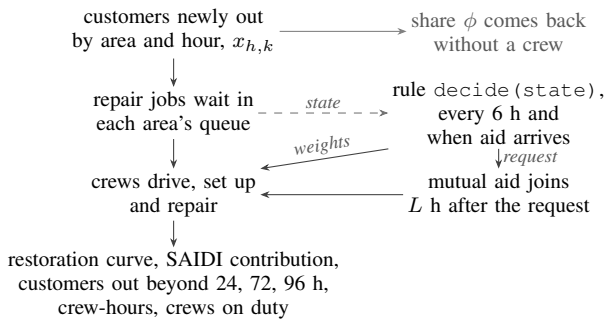


Fig. 2. The restoration loop. Damage replayed from the outage records becomes jobs, a rule decides where crews go and when to ask for aid, and the engine reports reliability outcomes.

TABLE I. THE STATE A RULE RECEIVES, PER AREA UNLESS NOTED

Field	Unit	Meaning
`t`	h	time since the storm window began (scalar)
`out`	customers	out, waiting for a crew or under repair
`waiting_jobs`	jobs	repair jobs in the queue
`working`	crews	working or driving to a job
`wait_h`	h	wait of the oldest queued job
`arrived, restored`	customers	outages so far; restored by crews so far
`crews, aid_crews`	crews	on duty with aid; brought by the request (scalars)
`forecast`	customers	interruptions the damage model expects
`customers`	customers	customers served
`dist_km`	km	distances between area centroids (matrix)
`counties`	table	name, state, centroid, customer density

Returns: weights $w_k \geq 0$, one per area, and an aid multiplier $m \geq 0$ (only the first request counts)

II. WHAT STORMCREWS DOES

A. Inputs

For each storm the engine needs the customers newly out in each area k during each hour h, written $x_{h,k}$, the customers served in each area and the distances between areas. The shipped data take $x_{h,k}$ from EAGLE-I, Oak Ridge National Laboratory's county record of customers out every 15 minutes [6], [7]. Each $x_{h,k}$ is the hourly rise in customers out, counted above the county's median of the 24 hours before the storm. Customers per county split each state's EAGLE-I customer count in proportion to Census housing units [8]. Weather (peak gust, rain and soil moisture) comes from ERA5 through Open-Meteo [9]–[11]. It enters only through the damage model behind the what-ifs and the damage forecast a rule may read. ERA5 covers the globe and the outage input is a plain area-by-hour count, so the same pipeline applies to any utility or regulator that keeps such records.

B. Restoration engine

The engine is a discrete-event simulation (Fig. 2). A share ϕ of $x_{h,k}$, measured per storm (9 % to 31 %), returns within 15 to 60 minutes without a crew, through reclosers, remote switching and quick manual switching. The rest becomes $\max\{1, \mathrm{round}((1-\phi)x_{h,k}/c)\}$ repair jobs with $c = 35$ customers per job on average, arriving at random within the hour. Job sizes are lognormal with dispersion σ and add up to the customers out. A crew that takes a job in area k coming from k' is busy for $d_{k'k}/60 + 0.5 + R_j$ hours, with centroid distance d in km and a lognormal repair time of mean $\bar{R}$ and coefficient of variation 0.6. Inside an area the largest job goes first by default. Oldest-first and a guard that serves any job waiting more than 48 h are options. Mutual aid requested with multiplier m at hour τ brings $\mathrm{round}(0.5\,N_0 m)$ crews at $\tau + L$, with N_0 the local fleet and a lag $L = 24$ h by default. If the rule has not asked by the time 1 % of the region's customers are out, the usual request ($m = 1$) goes out. Aid can also be switched off, and its lag and size changed.

C. What a rule sees and returns

At each decision point, every 6 hours and when aid arrives, a rule receives the state of Table I. It returns non-negative weights w_k, one per area, and optionally an aid multiplier. No field is normalized: all are in customers, crews, hours or km, and a rule that wants shares divides by customers served or by crews itself. The engine turns the weights into targets $T_k = N(t)\,w_k / \sum_i w_i$, with $N(t)$ the crews on duty. A crew freed in an area stays while the area is below its target. Otherwise it moves to the area with the largest deficit in crews or, with proportional dispatch, the largest deficit relative to its target. A rule may also set the job order and the dispatch mode. The default rule, current practice, weighs areas by their customers out.

D. Outputs in reliability terms

Every run returns the restoration curve and, from the outage d_j of each job j with c_j customers, the customer-hours $\sum_j c_j d_j$. Divided by the customers served and expressed in minutes, they give the storm's contribution to SAIDI as defined in IEEE Std 1366, with major event days included [12]. Divided by the customers interrupted they give the event's CAIDI. A run also counts customers out beyond 24, 72 and 96 hours and reports the time to restore the last customer, the worst area's mean outage, paid crew-hours (local crews from the start of the storm and aid crews from the request, both to the end of restoration), crew-hours on jobs and on the road, and the crews on duty once aid has joined.

E. Interfaces

StormCrews runs from Python, from a command line and from YAML scenario files, and an interactive notebook lets a user without code pick a storm, move the what-if and crew settings and compare the run with a pinned baseline. The repository documents every interface, scenario key and metric.

III. WHAT IT IS FOR AND WHAT IT IS NOT FOR

StormCrews answers regional questions on real storms (Table II), and its simplifications set the limits. Each utility territory keeps its share of the regional fleet and its own aid, and its crews serve only its counties, with every county assigned whole to one territory. Crews drive between area centroids. There is no feeder topology, no switching and no power flow. The damage of a real storm is replayed, not generated, except through the what-if model, which is

TABLE II. QUESTIONS STORMCREWS CAN AND CANNOT ANSWER

Fit for	Not fit for
How the SAIDI contribution and long outages of past storms respond to fleet size, aid lag and restoration rules	Dispatching or routing one utility's crews in a live storm
Weather what-ifs on real storms (stronger gusts, wetter soil) at regional scale	Switching sequences, feeder reconfiguration, power flow, voltage or protection
Comparing rules on a common, reproducible set of storms (researchers, regulators)	Estimated times of restoration for customers or feeders
Checking restoration standards and long-outage counts a regulator may set	Outcomes inside an area or split by utility territory

associational. Rules that request aid early read the hour of the peak from the data.

IV. CALIBRATION AND VALIDATION

Each storm has three parameters, the local fleet N_0, the mean repair time $\bar{R}$ and the job-size dispersion σ. In the anchored calibration used here, N_0 is set so that the local fleet plus the usual aid equals 4.0 crews per 1,000 customers out at the observed peak. That is the median of five public filings from New England and Pennsylvania that report line crews at work together with the same utility's peak (1.6 to 5.9 per 1,000; the Pennsylvania report counts line workers, taken at two per crew [13]). The median fleet is 342 crews in New England and 302 in Pennsylvania and West Virginia. $\bar{R}$ is then fitted so that current practice reproduces the observed customer-hours, and σ so that the curve matches the hours from the peak to one half and one tenth of it. Customer-hours and the two decay times are therefore targets. The peak is not, so it checks the fit, and it runs low under every calibration (0.88 to 0.94 of the observed, median over storms).

Because that fit is in sample, every storm was also predicted with parameters from the others. Leaving each storm out, $\log \bar{R}$ and σ are regressed, over the other storms of its region, on the log of the peak customers out and on the hours over which the middle 80 % of the damage arrives, both inputs of the engine, while the fleet keeps the anchor. The two predictors were chosen by this same test among seven candidate sets built from the inputs; the peak alone gives median errors of 28 and 24 %. Table III and Fig. 3 compare the two. Out of sample, a storm's customer-hours miss by 16 % in New England and 17 % in Pennsylvania and West Virginia at the median, and fall within 25 % for 67 % and 59 % of the storms. The hours from the peak to a tenth of it miss by 0 and 3 hours at the median. Summed over each region, held-out customer-hours come to 1.13 and 1.04 of those observed. The simulator replays observed storms closely and predicts an unseen storm only roughly.

The county pattern shows why crews stay in their territories. In Isaias, Connecticut lost 33.3 customer-hours per customer and the rest of New England 2.1. The simulation gives 28.5 and 3.7 (Fig. 4, Spearman 0.94 across counties); one pooled fleet for the whole region would give 22.0 and 5.7, spreading the long outages more evenly than the utilities, each restoring its own territory, did.

TABLE III. CURRENT PRACTICE AGAINST EVERY OBSERVED STORM, ANCHORED CALIBRATION, MEAN OF SEEDS SEEDS

Region	Parameters	Customer-hours (targeted)			Peak	Decay error (h)	
		sim/obs median	error median	within 25 %	sim/obs median	to 50 % median	to 10 % median
NE (39)	own storm	1.01	1 %	100 %	0.83	3	1
NE (39)	held out	0.91	16 %	67 %	0.79	3	0
PA–WV (46)	own storm	1.00	1 %	100 %	0.91	2	1
PA–WV (46)	held out	0.97	17 %	59 %	0.91	4	3

TABLE IV. RESTORATION RULES, 39 NEW ENGLAND STORMS, SUMS OVER STORMS

Rule	SAIDI	Customers out beyond			Worst	Travel
	(min)	24 h (M)	72 h (k)	96 h (k)	area (h)	(k crew-h)
Largest first	2,272	4.04	614	295	25.8	92
Oldest first	4,885	8.25	2,880	1,688	52.8	79
Proportional dispatch	2,271	4.06	614	295	22.2	90

Worst area: mean outage of the worst county, averaged over storms. Mean of 5 seeds.

V. ILLUSTRATIONS

Each illustration below is one command. It runs current practice on the 39 New England storms with the anchored calibration and 5 seeds. Summed over those storms, the simulated SAIDI contribution is 2,272 minutes, and 614 thousand customers stay out beyond 72 hours.

Weather. Raising the peak gust by 5 m/s in every county, through the damage model, multiplies the customers interrupted by 2.0, the SAIDI contribution by 3.3 and the customers out beyond 96 hours by 7.7 (Fig. 5a). The same crews face a longer queue. The median storm takes 173 hours to restore instead of 137, and paid crew-hours grow 66 %.

Crew resources. Across the range the filings report, 1.6 to 5.9 crews per 1,000 customers out at the peak, the SAIDI contribution goes from 97 % above to 21 % below its anchored value (Fig. 5b). Returns diminish. Going from 4.0 to 5.9 crews per 1,000 cuts SAIDI by 21 % and the customers out beyond 72 hours by 58 %, but it takes 20 % more paid crew-hours and 24 % more driving. Aid that arrives 12 hours after the request instead of 24 changes SAIDI by -13 %, and a 48-hour lag by +18 % (Fig. 5c).

Restoration rules. Table IV compares job orders and dispatch modes. Serving the oldest job first instead of the largest raises SAIDI by 115.0 % and lengthens the tail too, because large faults wait behind small ones, even though crews drive 13.8 % less. Proportional dispatch leaves SAIDI almost unchanged (-0.1 %) and shortens the worst county's mean outage by 14 %. A crew freed in a busy area then goes to the area furthest behind its own target rather than the one short of most crews.

Assumptions. The figure of 35 customers per job is an assumption. It lies between the mean of about 31 customers per outage at the peak of the five storms in [5] and the median of about 38 over 55 Connecticut storms read from Fig. 4 of [4]. Table V refits $\bar{R}$ at 20 and 55 customers per job and reruns the illustrations. The relative effects move by a few points, while the long-tail counts and crew-hours change much more, so those depend on the assumption. Moving the reactive trigger

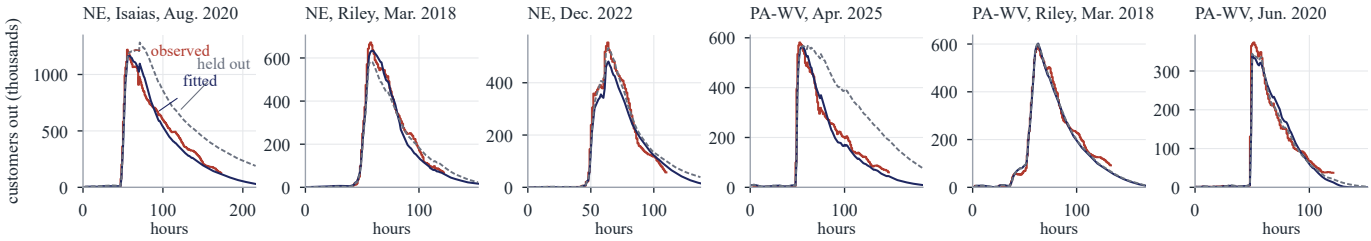


Fig. 3. The fitted simulator follows observed restoration, and parameters borrowed from other storms follow it less closely. Observed EAGLE-I customers out (red), current practice with the storm's own anchored parameters (navy) and with repair time and dispersion predicted from the other storms of its region (gray, dashed); six storms, seed 0.

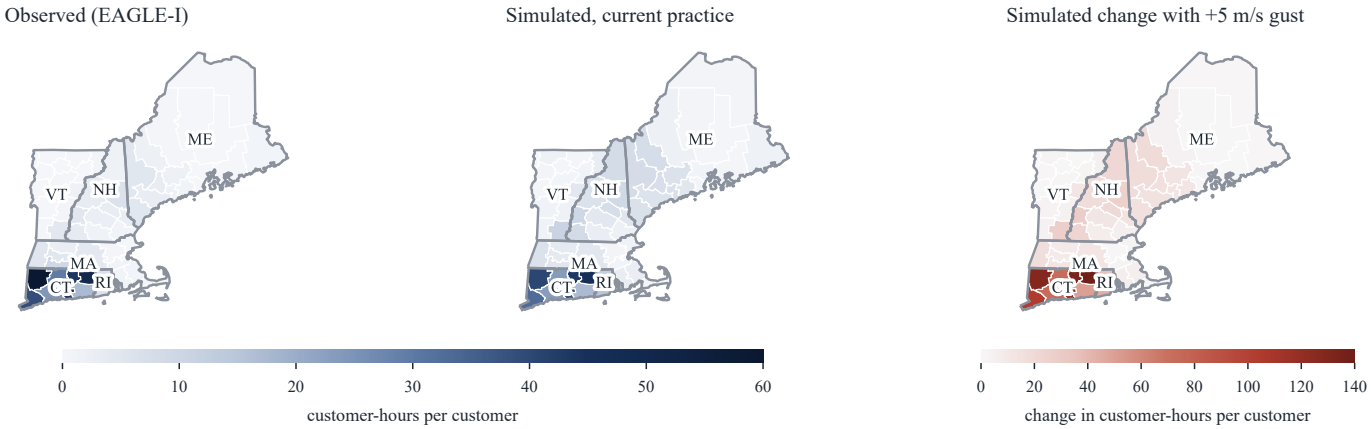


Fig. 4. The simulator places a real storm's outages where they happened. Tropical Storm Isaias by county: observed customer-hours per customer (EAGLE-I), the same simulated under current practice with the anchored calibration (seed 1), and the simulated change if peak gusts had been 5 m/s stronger.

TABLE V. SENSITIVITY TO THE CUSTOMERS BEHIND ONE REPAIR JOB

Customers	Refitted	SAIDI change against the real storms			Out >96 h	On jobs
per job	$\bar{R}$ (h)	gust +5	75 % fleet	48 h aid lag	(k)	(M crew-h)
20	1.6	×3.52	+24 %	+18 %	368	3.14
35	2.6	×3.34	+21 %	+18 %	295	2.59
55	3.4	×3.18	+19 %	+17 %	224	2.17

from 0.5 % to 2 % of customers out changes SAIDI by −3.8 % and +1.1 %.

VI. CONTRIBUTION AGAINST EXISTING TOOLS

Table VI compares StormCrews with the closest tools. Walsh et al. model Connecticut crews and mutual-aid crews on five storms with a utility's data [5]. Oak Ridge's RePOWERD work routes crews over damaged roads on hypothetical outages [14], [15], and Willems et al. estimate the restoration times of five hurricanes statistically [16]. Open platforms are reproducible but do not replay storms with crews and aid. InfraRisk sends repair crews over the road network of test cities [17], PowerModelsRestoration orders restoration by optimization [18] and GridLAB-D simulates feeders [19]. The EAGLE-I outage records give real storms without decisions [6]. StormCrews brings these pieces together: real storms calibrated from public records, crews with travel, mutual aid with its lag, rules a user writes, and results in reliability indices, all open and checked by 59 tests.

TABLE VI. STORMCREWS AND RELATED TOOLS (● YES, ○ PARTLY, – NO)

Tool	Real storms	Public data	Crews, travel	Mutual aid	Open, reproducible
StormCrews	●	●	●	●	●
Walsh et al. [5]	●	–	●	●	○
RePOWERD [14], [15]	○	○	●	○	–
Willems et al. [16]	●	●	–	–	–
InfraRisk [17]	–	●	●	–	●
PowerModelsRestoration [18]	–	●	–	–	●
GridLAB-D [19]	–	●	–	–	●
EAGLE-I outage data [6]	●	○	–	–	●

VII. AVAILABILITY

The code and data are at https://github.com/abrahambohorquez/stormcrews (version 0.3.0, Python 3.10 or later). It installs with pip from GitHub, or from a clone with `pip install -e .` run in its folder. A run takes a scenario, a storm, a rule and a seed, and each result reported here comes from one command:

```
pip install \
  git+https://github.com/abrahambohorquez/stormcrews

from stormcrews import Simulator
from stormcrews.rules import CurrentPractice
sim = Simulator("new_england", calibration="anchored")
r = sim.run(CurrentPractice(), storm="isaias", seed=1)
print(r.metrics["saidi_min"], r.metrics["over_72h"])

python paper_gtdla/experiments.py all
python paper_gtdla/make_figures.py   # every figure
python paper_gtdla/make_macros.py    # every number
```

The code is under the MIT license and the derived data

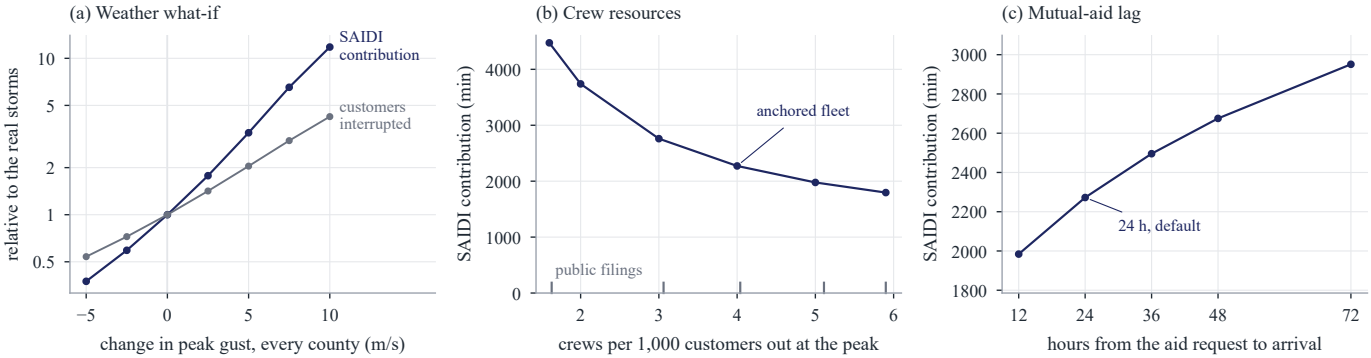


Fig. 5. Stronger gusts add outage minutes faster than interruptions, and more crews help with diminishing returns. Simulated current practice on the 39 New England storms, anchored calibration, mean of 5 seeds. (a) Peak gust changed in every county. (b) Local fleet from 1.6 to 5.9 crews per 1,000 customers out at the peak, the range of the public filings (ticks). (c) Hours from the aid request to arrival.

under CC BY 4.0, with the attributions of the repository's data license. The outage data are from EAGLE-I, Oak Ridge National Laboratory, under CC BY 4.0 [7]. The weather contains modified Copernicus Climate Change Service information, retrieved through Open-Meteo (weather data by Open-Meteo.com) under CC BY 4.0 [10], [11]. Neither Oak Ridge National Laboratory, the U.S. Department of Energy nor ECMWF endorses StormCrews.

VIII. Conclusion

StormCrews turns public outage records into a restoration simulator that a distribution planner, a regulator or a researcher can rerun and change. On 85 real storms it shows what a rule, a fleet size or a stronger storm does to the SAIDI contribution, to the customers out for days and to the crew-hours spent. It replays observed storms closely and predicts unseen ones roughly. It works at county level with each utility's share of the crews, so it serves regional what-ifs and the comparison of rules, not the dispatch of one utility. A third region comes next.

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